Band-1 Computational Experiment: van der Waerden — Finite Exact Numbers Ledger

Human + ZoraASI Band-1 (executor)

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Band-1 Computational Experiment: van der Waerden — Finite Exact Numbers + Box Certificates

Honesty stamp. This note does **not** claim the van der Waerden asymptotic problem, Szemerédi's theorem quantitative bounds, or any Clay/Millennium problem is solved. It does **not** claim a finished Theory of Everything. **GATE-04 collection remains LOCKED** (no invented J; no unlock). Band-1 here is a **literature ledger** of known finite exacts $W(k;r)$ plus **box certificates for small values only**. Full asymptotics and the next open exacts ($W(7;2)$, $W(5;3)$, $W(4;4)$, …) remain **OPEN**. No RH / P≠NP / YM / Frankl / Dedekind-$M(9)$ / full ES / full Hadamard "solved'' claims appear in this note.

Abstract

Let $W(k;r)$ be the least n such that every r-colouring of $\{1,\dots,n\}$ contains a monochromatic arithmetic progression of length k. We (i) ledger classical exact values surveyed 2026-09-11; (ii) certify on-box $W(3;2)=9$, $W(4;2)=35$, and $W(3;3)=27$ by pruned exhaustive search (lower witnesses + upper emptiness proofs); (iii) name the next notable open exacts from the literature without inventing new values. Asymptotic status remains **OPEN**.

Problem statement

Van der Waerden (1927): for every $k \geq 1$ and $r \geq 1$ there exists a finite $W(k;r)$ as above. Computing exact $W(k;r)$ is a classical hard combinatorial-search problem; only a sparse table of exacts is known.

Notation in this artifact: $W(k;r)$ (length; colours). Equivalent classical (r,k)-order writes $W(r,k)$ with the same numeric table ($W(3;2)=W(2,3)=9$, etc.).

Literature ledger (survey 2026-09-11; not re-proved except small box cases)

$W(k;r)$	Value	Status	Source
$W(3;2)$	**9**	exact; **box-certified tonight**	classical / Chvátal 1970; OEIS A005346
$W(4;2)$	**35**	exact; **box-certified tonight**	classical / Chvátal 1970; OEIS A005346

$W(k;r)$	Value	Status	Source
$W(5;2)$	**178**	exact (literature)	Stevens–Shantaram 1978
$W(6;2)$	**1132**	exact (literature)	Kouril–Paul 2008 Exp. Math.
$W(3;3)$	**27**	exact; **box-certified tonight**	Chvátal 1970
$W(3;4)$	**76**	exact (literature)	Beeler–O'Neil 1979
$W(4;3)$	**293**	exact (literature)	Kouril 2012 Integers A46
$W(7;2)$	—	**OPEN** (lower > 3703)	Wikipedia / OEIS tables 2026
$W(5;3)$	—	**OPEN** (lower > 2173)	same
$W(4;4)$	—	**OPEN** (lower > 1048)	same

No new exact values are invented in this note. Large literature exacts above are cited, not re-proved.

Method (box)

1. **Definition check.** Colouring $c : \{1, \dots, n\} \to \{0, \dots, r-1\}$; mono k-AP means $c(a) = c(a+d) = \cdots = c(a+(k-1)d)$ for some $a, d \geq 1$.
2. **Incremental prune.** DFS colouring; after placing colour at n, only test k-APs that **end** at n (standard filter).
3. **Lower bound.** Exhibit one k-AP-free r-colouring of $[W-1]$.
4. **Upper bound.** Prove by exhaustive pruned search that every r-colouring of $[W]$ has a mono k-AP (equivalently: zero free colourings).
5. **Fail-closed.** Time budgets on larger cases; if incomplete, report partial + literature honestly rather than hang or invent.

Code: `/workspace/experiments/band1/vandervaerden_finite/vdw_exact.py`.

Results (numbers!)

Unit tests

`PASS` (mono-AP detector; classic witness `RRBBRRBB` AP3-free on [8]).

Box-certified exacts

Claim	Lower witness ($[W-1]$)	Upper	Wall time (approx.)
$W(3;2) = 9$	`RRBBRRBB` (and **6** free 2-colourings on [8]; **0** on [9])	proved none free on [9]	< 0.01 s

Claim	Lower witness ($[W-1]$)	Upper	Wall time (approx.)
$W(4;2) = 35$	`RRBRRRBBBRBRRBRRRBBBRBRRBRRRBBBRBB` on [34]	proved none free on [35]	~ 0.14 s
$W(3;3) = 27$	`001100121220200101120221 21` on [26]	proved none free on [27]	~ 4.4 s

All three certificates completed on this box under the Band-1 time budget (< 5 min total).

Artifacts: `results.json`, `literature_ledger.json`, `SUMMARY.txt`.

Literature next open (not solved here)

- $W(7;2)$ — first open 2-colour length-7 exact (literature lower > 3703).
- $W(5;3)$ — open 3-colour length-5 (literature lower > 2173).
- $W(4;4)$ — open 4-colour length-4 (literature lower > 1048).

Why Band-1 vs asymptotics

- **Band-1 (this note):** finite exact ledger + small exhaustive certificates; incremental frontier.
- **Not claimed:** Gowers / Shelah quantitative van der Waerden bounds settled; Szemerédi rates; any new exact beyond the box certificates above; full diagonal table closed.

Non-claims (explicit)

- No Clay-closed stamp.
- No finished-ToE claim.
- GATE-04 collection **LOCKED**.
- Full van der Waerden asymptotics **OPEN — not solved**.
- $W(7;2)$, $W(5;3)$, $W(4;4)$ remain literature-OPEN.

Reproducibility

- Code: `/workspace/experiments/band1/vandervaerden_finite/`
- Run: `python3 vdw_exact.py`
- Log: `/workspace/experiments/logs/2026-09-11_vandervaerden_finite.log`
- Artifacts: `results.json`, `literature_ledger.json`, `SUMMARY.txt`

Conclusion

Box: certified $W(3;2) = 9$, $W(4;2) = 35$, $W(3;3) = 27$ by lower witnesses + upper emptiness proofs.

Literature: $W(5;2) = 178$, $W(6;2) = 1132$, $W(3;4) = 76$, $W(4;3) = 293$ cited as known exacts; next notable open exacts $W(7;2)$, $W(5;3)$, $W(4;4)$.

Asymptotic / full table: OPEN — not claimed solved. GATE-04 LOCKED.